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AI speech-based emotion recognition

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Abstract

This report documents the complete two-semester graduation project Real-Time Multi-modal Speech & Face Emotion Recognition System (ENCS5300), building on the offline machine-learning baseline established in the first semester [1].

The first semester validated a traditional machine-learning approach using handcrafted acoustic features and an SVM–Random Forest Voting Classifier ensemble, achieving 89.49% accuracy on five RAVDESS [4] emotions and 83.68% on all eight emotions. A parallel deep-learning investigation using a hierarchical CNN-BiLSTM architecture and the Wav2Vec2 transformer model was also conducted.

This semester delivers four primary contributions. First, the trained ensemble is deployed in a fully operational real-time application combining continuous microphone-captured speech with optional facial expression analysis from a webcam, producing second-by-second predictions expressed as emotion pairs—the two simultaneously dominant emotions drawn from all 28 canonical combinations. Second, a three-stage audio preprocessing pipeline (noise gate, pre-emphasis, VAD [7]) is integrated, reducing spurious predictions by 27.6% in naturalistic sessions. Third, a probability calibration layer corrects systematic dataset-specific biases without retraining. Fourth, a research-grade reporting module exports timestamped session data as CSV, Excel, and PDF documents with embedded statistical charts for quantitative modality-contribution analysis.

To evaluate generalization, the system was tested on an unseen out-of-distribution corpus—TESS. The results showed that while top-1 accuracy decreased on out-of-distribution samples, the pair-based representation consistently improved the overall prediction quality by capturing secondary emotional information.

This demonstrates the effectiveness of the proposed pairwise framework in handling real-world scenarios where emotions may not match the training distribution.

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List of Symbols and Observations

Symbol	Context	Description
E_{rms}	Equation (2.1)	Root Mean Square energy of an audio buffer frame.
τ_{gate}	0.01	Energy threshold limit utilized by the audio Noise Gate pipeline stage.
α	0.97	First-order high-pass Finite Impulse Response (FIR) pre-emphasis filter coefficient.
θ_{VAD}	0.25	Minimum voice activity frame ratio required to validate speech presence.
$\hat{\mathbf{p}}$	$\mathbb{R}^8$	Initial output raw joint probability vector generated across the 8 emotion classes.
τ_{min}	0.10	Minimum confidence boundary cut-off used prior to parsing secondary emotion pairs.
w_v	0.55	Relative weight factor assigned to the voice modality stream during late decision fusion.
Δ	Performance Metric	Confidence variance delta evaluated between fused modality output and voice-only scores.

Chapter 1

Introduction

1.1 Project Overview and Continuation

Speech Emotion Recognition (SER) aims to automatically identify emotional states from voice characteristics, enabling machines to better understand and respond to human needs [8]. This graduation project investigates SER under realistic, low-data, and real-time conditions across two semesters.

The first semester [1] established two complementary approaches: (i) a traditional machine-learning baseline using handcrafted acoustic features—MFCCs, chroma, mel-spectrogram, zero-crossing rate, and RMS energy—with an SVM–Random Forest ensemble achieving 83.68% on all eight RAVDESS emotions and 89.49% on five emotions; and (ii) a frequency-based clustering approach using CNN-BiLSTM models achieving 56% on the same task. The key finding was that traditional ML with well-engineered features significantly outperforms deep learning under data scarcity.

The second semester addresses the critical gap: the transition from a static offline classifier to a deployable, real-time, multimodal system. The objectives are:

1. Deploy the trained SVM–RF ensemble in a live application with sub-second inference latency.
2. Implement a robust audio preprocessing pipeline improving prediction quality on naturalistic speech [7].
3. Extend the output from single-emotion labels to emotion pairs, capturing the co-occurrence of emotional states [8].
4. Integrate facial expression analysis via DeepFace [11] and fuse it with voice predictions using weighted probability averaging [3].
5. Build a research-grade reporting module for quantitative session analysis.
6. Formally validate the deployed pipeline on in-distribution (RAVDESS) data and an out-of-distribution corpus (TESS).

1.2 Motivation for Emotion Pairs

Classical SER systems output a single discrete label, implicitly assuming mutually exclusive emotional states. In practice, emotions co-occur: a speaker may be simultaneously happy and surprised, or sad and calm [8]. The RAVDESS dataset itself contains emotions with significant acoustic overlap, as demonstrated by confusion patterns in the first semester [1]. Rather than forcing a single-label decision, this system outputs the top-2 emotions by probability, forming one of 28 canonical pairs. This representation aligns with psychological dimensional models of emotion where arousal and valence jointly determine emotional state [8], and is practically informative for applications such as mental health monitoring [10].

1.3 Report Structure

Chapter 2 provides the theoretical background. Chapter 3 describes the system architecture. Chapter 4 details the application interface. Chapter 5 covers the reporting module. Chapter 6 presents all experimental results including cross-corpus validation. Chapter 7 discusses findings, limitations, and comparisons with related work. Chapter 8 concludes with a summary and future directions.

Chapter 2

Theoretical Background

2.1 Audio Preprocessing for Real-Time SER

Unlike offline classification on clean studio-quality recordings, a real-time system must handle continuous microphone input containing noise, silence, and varying loudness. Research confirms that preprocessing steps significantly improve SER accuracy on naturalistic speech [6, 7]. Three stages are applied sequentially.

2.1.1 Noise Gate

A noise gate suppresses frames below an energy threshold, preventing spurious predictions during silence or ambient noise. The root mean square (RMS) energy is computed over the current 3-second audio buffer:

$$E_{\text{rms}} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} x[n]^2} \quad (2.1)$$

If $E_{\text{rms}} < \tau_{\text{gate}} = 0.01$, the buffer is classified as non-speech and inference is skipped. The threshold was determined empirically on a standard laptop microphone in an office environment.

2.1.2 Pre-emphasis Filter

Speech signals exhibit spectral tilt: low frequencies carry more energy than high frequencies, under-representing consonants and fricatives that carry emotional cues [1, 5]. A first-order high-pass FIR filter corrects this:

$$y[n] = x[n] - \alpha \cdot x[n-1], \quad \alpha = 0.97 \quad (2.2)$$

The coefficient $\alpha = 0.97$ follows the standard speech processing convention [5].

2.1.3 Voice Activity Detection

Voice Activity Detection (VAD) classifies individual frames as speech or non-speech, ensuring that only valid speech segments enter the classification pipeline [7]. Two implementations are provided:

WebRTC VAD (primary): Google’s industry-standard algorithm operating on 10, 20, or 30 ms frames at 16 kHz, configured at aggressiveness level 2.

Energy-based VAD (fallback): When `webrtcvad` is unavailable, per-frame RMS energy is compared to an adaptive threshold. A buffer is accepted as speech if at least 25% of frames exceed 50% of the mean frame energy:

$$\text{speech_ratio} = \frac{\#\{f : E_{\text{rms}}[f] > 0.5 \bar{E}\}}{\text{total frames}} \geq \theta_{\text{VAD}} = 0.25 \quad (2.3)$$

2.2 Feature Extraction Pipeline

The feature extractor replicates precisely the pipeline used during training to guarantee train/inference consistency [1]. A 162-dimensional feature vector is extracted from each 3-second audio window (Table 2.1).

Table 2.1: Feature extraction pipeline(dimensions and acoustic motivation)

Feature Group	Coefficients	Statistics	Dims
MFCCs	40	Mean + Std per coefficient	80
Chroma	12	Mean per pitch class	12
Mel-spectrogram	64	Mean (dB) per band	64
Zero Crossing Rate	1	Mean + Std	2
RMS Energy	1	Mean + Std	2
Total			162

MFCCs provide a compact spectral representation matching human auditory perception and are the most widely used SER features [5, 6]. Pitch is a primary emotion indicator: high-arousal emotions exhibit elevated mean pitch and greater variability, while low-arousal emotions show reduced pitch [9].

2.2.1 Feature Selection and Scaling Order

The 162-dimensional vector is reduced to 80 dimensions using `SelectKBest` with the ANOVA F-statistic, then normalised by `StandardScaler`. The selector is applied *before* the scaler—an essential ordering constraint, since the scaler was fitted on the already-

reduced 80-dimensional space during training [1]:

$$\mathbf{f}_{\text{raw}} \in \mathbb{R}^{162} \xrightarrow{\text{SelectKBest}(k=80)} \mathbf{f}_{\text{sel}} \in \mathbb{R}^{80} \xrightarrow{\text{StandardScaler}} \mathbf{f}_{\text{norm}} \in \mathbb{R}^{80} \xrightarrow{\text{VotingClassifier}} \hat{\mathbf{p}} \in \mathbb{R}^8 \quad (2.4)$$

2.3 Emotion Pair Detection

2.3.1 Motivation

Emotions are rarely mutually exclusive in natural speech [8]. The ensemble outputs a probability vector $\hat{\mathbf{p}} = [p_1, \dots, p_8]^\top$ summing to 1 over the eight RAVDESS emotion classes [4]. Rather than taking $\arg \max$, the system identifies the top-2 emotions:

$$e_1 = \arg \max_i p_i \quad (2.5)$$

$$e_2 = \arg \max_{i \neq e_1} p_i \quad (2.6)$$

The pair (e_1, e_2) is mapped to a canonical ordering over all $\binom{8}{2} = 28$ combinations of $\mathcal{E} = \{\text{neutral, calm, happy, sad, angry, fear, disgust, surprise}\}$.

2.3.2 Pair Confidence Score

Each pair is scored as the sum of its constituent probabilities, then normalised across all 28 pairs:

$$s(e_i, e_j) = p_i + p_j, \quad \tilde{s}(e_i, e_j) = \frac{s(e_i, e_j)}{\sum_{(a,b) \in \mathcal{P}} s(a, b)} \quad (2.7)$$

2.4 Probability Calibration

2.4.1 The Disgust Over-Prediction Problem

A systematic bias was observed during real-speech testing: “disgust” appeared as the top prediction in 64.3% of frames during naturalistic conversational speech. This is a documented artefact of models trained exclusively on acted emotional speech [6]: disgust in RAVDESS is performed with a low, tense, creaky voice whose acoustic profile closely resembles ambient silence and non-emotional baseline speech captured by a typical laptop microphone.

2.4.2 Calibration Weights

To correct this bias without retraining, a per-class multiplicative calibration layer is applied to the raw softmax probabilities:

$$p'_i = p_i \cdot w_i, \quad \hat{p}'_i = \frac{p'_i}{\sum_j p'_j} \quad (2.8)$$

Table 2.2 shows the weights derived from empirical observation.

Table 2.2: Manual probability calibration weights per emotion class

Emotion	Weight w_i	Rationale
neutral	1.20	Under-predicted on real speech
calm	1.10	Under-predicted on real speech
happy	1.00	No systematic bias
sad	1.00	No systematic bias
angry	0.90	Slight over-prediction
fear	0.90	Slight over-prediction
disgust	0.40	Substantial reduction — main bias correction
surprise	0.95	

2.4.3 Minimum Confidence Threshold

After calibration and renormalisation, any emotion below $\tau_{\min} = 0.10$ (10%) is zeroed out before pair selection:

$$\hat{p}''_i = \begin{cases} \hat{p}'_i & \text{if } \hat{p}'_i \geq 0.10 \\ 0 & \text{otherwise} \end{cases} \quad (2.9)$$

2.5 Multimodal Fusion: Voice + Face

2.5.1 Facial Emotion Analysis

Following the multimodal framework of Subbulakshmi et al. [3], the DeepFace library [11] is integrated to provide face-based emotion probabilities. DeepFace outputs a 7-class distribution over $\mathcal{E}_{\text{face}} = \{\text{angry, disgust, fear, happy, neutral, sad, surprise}\}$, which differs from the voice model’s 8-class set by the absence of “calm.”

2.5.2 Emotion Space Mapping

“Calm” has no face equivalent. It is interpolated from face “neutral” following the arousal-based relationship established in the first semester [1]: calm is acoustically similar to

neutral but at even lower arousal, so face “neutral” probability is split:

$$p_{\text{neutral}}^{\text{face}} \longrightarrow \begin{cases} p_{\text{neutral}}^{\text{face}} = 0.60 \cdot p_{\text{neutral}}^{\text{face}} \\ p_{\text{calm}}^{\text{face}} = 0.40 \cdot p_{\text{neutral}}^{\text{face}} \end{cases} \quad (2.10)$$

2.5.3 Weighted Probability Fusion

The fused probability vector is a weighted average of calibrated voice probabilities and mapped face probabilities [3]:

$$p_i^{\text{fused}} = w_v \cdot p_i^v + (1 - w_v) \cdot p_i^f, \quad w_v = 0.55 \quad (2.11)$$

The voice weight $w_v = 0.55$ is set slightly above 0.5 because the voice model is a custom-trained ensemble on RAVDESS [4] achieving 83.68% accuracy, while the face model is a general-purpose pretrained CNN. When only one modality is available, the system falls back to that modality’s predictions without fusion.

2.6 Deep Learning Speech Emotion Recognition

2.6.1 Motivation for Deep Learning

Although the traditional machine-learning ensemble achieved strong performance on the RAVDESS dataset [4], a deep-learning approach was explored to investigate whether modern transformer-based speech models could learn richer emotional representations directly from raw audio signals without relying entirely on handcrafted features.

Traditional SER systems depend heavily on manually extracted acoustic features such as MFCCs, pitch, energy, and spectral statistics [5, 6]. While these features are effective under limited-data conditions, they may fail to capture higher-level temporal and contextual emotional patterns present in speech. For this reason, a transfer-learning approach using the pretrained Wav2Vec2 model was investigated. Wav2Vec2 is a self-supervised transformer-based speech representation model that learns contextual speech embeddings from raw waveforms.

2.6.2 Hierarchical Deep Learning Approach

Before adopting Wav2Vec2, a CNN-BiLSTM hierarchical architecture was explored following the methodology proposed by Barhoumi and BenAyed [2]. The goal was to reduce confusion between acoustically similar emotions by first separating emotions into high-arousal and low-arousal groups. The hierarchical structure consisted of two stages:

1. **Arousal Classification Stage:** Speech was first classified into high-arousal emotions {angry, fearful, disgust, surprised} and low-arousal emotions {neutral, calm, sad, happy}.
2. **Specialized Emotion Classification Stage:** Separate CNN-BiLSTM classifiers were trained for each arousal group.

The motivation was that emotions within the same arousal level share similar spectral and prosodic characteristics [9]. The hierarchical model achieved approximately 56% accuracy on the 8-emotion RAVDESS classification task. While this slightly improved on the flat CNN-BiLSTM baseline, the model still suffered from instability and limited generalization due to the relatively small dataset size.

2.6.3 Wav2Vec2 Architecture

Wav2Vec2 is a transformer-based deep-learning architecture developed by Facebook AI for speech representation learning, pretrained using self-supervised learning on large amounts of unlabeled speech audio. The architecture consists of three main components:

1. **Convolutional Feature Encoder:** The raw waveform is processed through a stack of convolutional layers, converting it into latent speech representations while reducing temporal resolution.
2. **Transformer Encoder:** The latent representations are processed using a transformer encoder containing multi-head self-attention layers, allowing the model to learn contextual relationships between distant speech segments.
3. **Classification Head:** A fully connected layer was added on top of the pretrained model to map learned representations to emotion classes.

2.6.4 Transfer Learning and Fine-Tuning

Although Wav2Vec2 is pretrained on large speech corpora, the original model is not designed specifically for emotion recognition. Therefore, transfer learning was used by fine-tuning the model on emotional speech data. Fine-tuning updates the pretrained weights using labeled emotional speech samples, allowing the model to adapt its representations toward emotional classification. The model was fine-tuned on the RAVDESS dataset [4] across all eight emotion classes: neutral, calm, happy, sad, angry, fearful, disgust, and surprised.

2.6.5 Preprocessing for Wav2Vec2

Unlike the traditional pipeline that requires explicit handcrafted feature extraction, Wav2Vec2 operates directly on the raw waveform. However, several preprocessing steps were still

required for consistent real-time inference: converting stereo audio to mono; resampling to 16 kHz (the native rate expected by Wav2Vec2); applying a pre-emphasis filter to strengthen high-frequency speech components; splitting speech into fixed-length chunks; and padding or truncating segments to ensure uniform tensor dimensions. The pre-emphasis filter is defined as:

$$y[n] = x[n] - \alpha x[n-1], \quad \alpha = 0.97 \quad (2.12)$$

This filter improves the representation of consonants and higher-frequency emotional cues [5].

2.6.6 Attention Mask Creation

Wav2Vec2 requires an attention mask to distinguish valid speech samples from padded samples inside each batch. The attention mask is a binary vector where $m_i = 1$ for a valid audio sample and $m_i = 0$ for a padding sample. This prevents the transformer self-attention mechanism from attending to artificial padding values that do not correspond to real speech.

2.6.7 Logits and Softmax Probabilities

The final output layer produces a vector of raw prediction scores called logits, $\mathbf{z} = [z_1, z_2, \dots, z_8]$. These are unconstrained real-valued confidence scores, not probabilities. To convert logits into normalized probabilities, the softmax function is applied:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^8 e^{z_j}} \quad (2.13)$$

such that $\sum_{i=1}^8 p_i = 1$. The emotion with the highest probability becomes the predicted class.

2.7 Pairwise Expert Models

To extend the system beyond single-emotion classification, we introduce a set of specialized binary classifiers referred to as **pairwise expert models**. Each model is trained to distinguish between two specific emotions (e.g., happy vs surprise, sad vs fearful).

The motivation behind this design is that certain emotions are closely related or often confused with each other. By training dedicated models on pairs of emotions, we aim to capture fine-grained differences that may not be fully learned by a general multi-class classifier.

Each pair model is based on the same Wav2Vec2 architecture, fine-tuned on a filtered subset of the dataset containing only the two target emotions. This allows the model to focus exclusively on learning the distinction between those emotions.

The pairwise models achieved high accuracy on their respective binary classification tasks, indicating their effectiveness in distinguishing between closely related emotional states.

In the proposed system, these models are not used to replace the general classifier, but rather to complement it. Specifically, they are used as a **verification layer** to evaluate the relationship between the primary and secondary emotions predicted by the general model.

This design enables the system to incorporate both global classification capabilities and specialized local decision-making, resulting in a more reliable two-emotion prediction framework. To further evaluate the behavior of these models in practice, we analyze their performance in detail in the following section.

2.7.1 Pairwise Emotion Modeling and Multi-Emotion Prediction

To better capture the complexity of human emotional expression, the system was extended beyond single-label classification to predict two emotions for each input.

Instead of relying only on a single dominant emotion, the proposed approach models both a primary and a secondary emotion, providing a more realistic representation of speech signals.

To support this design, a pairwise modeling approach was introduced. Multiple specialized models were trained, each focusing on distinguishing between a specific pair of emotions.

Each pair model was implemented as a three-class classifier, where the two target emotions are explicitly modeled, and all remaining emotions are grouped into an “other” class.

This design allows the model not only to distinguish between the two target emotions, but also to reject unrelated emotional inputs, making it more robust for real-world scenarios.

2.7.2 Real-Time Chunk-Based Emotion Analysis

A real-time speech emotion analysis system was implemented using chunk-based inference. Continuous microphone audio is divided into short overlapping segments called chunks, each processed independently through the Wav2Vec2 model. For every chunk, the system outputs individual emotion probabilities, the most dominant emotion, and the top-2 emotion pair. Chunk-based processing improves temporal responsiveness and allows the system to monitor emotion changes continuously during speech.

2.7.3 Multi-Emotion Probability Distribution and Pair Detection

Instead of producing only a single emotion label, the system displays the full probability distribution across all emotions, which is more informative because emotions are not always mutually exclusive [8]. The top-2 emotions were combined into emotion pairs; since there are 8 emotion classes, the system generates $\binom{8}{2} = 28$ possible pairs. A mixed-emotion interpretation layer was also implemented: if the top two emotions have similar probabilities, the speech is interpreted as a mixed emotional state, while semantically contradictory combinations (e.g., happy + sad) were filtered to avoid unrealistic outputs.

2.7.4 Challenges Encountered

Several challenges were observed during real-time deployment of the deep learning model:

- **Fearful over-prediction:** The model frequently predicted fearful emotion during natural speech due to domain mismatch between acted emotional speech and real conversational speech.
- **Disgust bias:** Disgust was strongly over-predicted because of similarities between low-energy speech and acted disgust samples in RAVDESS.
- **Latency constraints:** Transformer inference is computationally heavier than traditional ML models, making real-time optimization necessary.
- **Dataset limitations:** RAVDESS is relatively small for transformer fine-tuning, which limited generalization.

These findings mirror the domain-mismatch challenges observed with the traditional ensemble (Section 6.1) and further motivate the calibration and preprocessing strategies adopted in the deployed system. Related deep architectures using multiple acoustic features with CNNs [13] and recurrent networks with local attention [14] report similar sensitivity to training-data scale.

2.7.5 Summary

The deep learning component explored both hierarchical CNN-BiLSTM architectures and transformer-based Wav2Vec2 models for SER. While traditional machine learning remained more stable under limited-data conditions, the transformer-based approach provided a more advanced contextual speech representation framework capable of real-time chunk-based emotion analysis, multi-emotion probability estimation, and mixed emotion interpretation. This comparative exploration expanded the research scope of the project and provided deeper insight into the strengths and limitations of each approach under limited-data real-time SER conditions.

Chapter 3

System Architecture

3.1 Overview

The application is a modular Python project deployed locally via the Gradio framework [12] at <http://localhost:7861>. The modular design separates audio processing, face analysis, fusion, and reporting into independently testable components, following software engineering best practices for maintainability.

3.2 Project File Structure

emotion_project/	
config.py	Central path & constant definitions
train.py	Full offline training pipeline
realtime_app.py	Gradio application entry point
validate.py	Raw model validation script
validate_pipeline.py	Full app pipeline validation
validate_tess.py	Cross-corpus TESS validation
src/	
feature_extractor.py	EnhancedAudioFeatureExtractor (162 features)
preprocessor.py	AudioPreprocessor (noise gate, pre-emph, VAD)
face_analyser.py	FaceEmotionAnalyser (DeepFace + OpenCV)
fusion.py	Weighted probability fusion module
reporter.py	SessionRecorder + CSV/Excel/PDF export
model/	
model.pkl	Trained VotingClassifier
scaler.pkl	Fitted StandardScaler (80 features)
selector.pkl	Fitted SelectKBest (162 to 80)
emotion_labels.json	Class index to label mapping
outputs/	Session export destination

3.3 Concurrent Thread Architecture

Real-time operation requires that audio capture, feature extraction, face analysis, and UI updates run concurrently without blocking each other. The system uses four concurrent threads (Figure 3.1):

1. **Audio callback thread** (`sounddevice`): Runs at OS scheduling priority; appends 100 ms audio chunks to a thread-safe circular buffer of capacity $44100 \times 3 = 132,300$ samples.
2. **Voice prediction thread**: Wakes every 1 second, snapshots the buffer, runs the full preprocessing $\rightarrow$ feature $\rightarrow$ model $\rightarrow$ calibration $\rightarrow$ pair selection pipeline, and writes the result to a lock-protected shared state.
3. **Face analysis thread**: Processes the most recent webcam frame through DeepFace [11] every 1 second, storing results in its own lock-protected state.
4. **Gradio timer thread**: A `gr.Timer` fires every 1 second, reads both shared states, performs fusion (Equation (2.11)), and pushes an HTML result card to the browser via WebSocket.

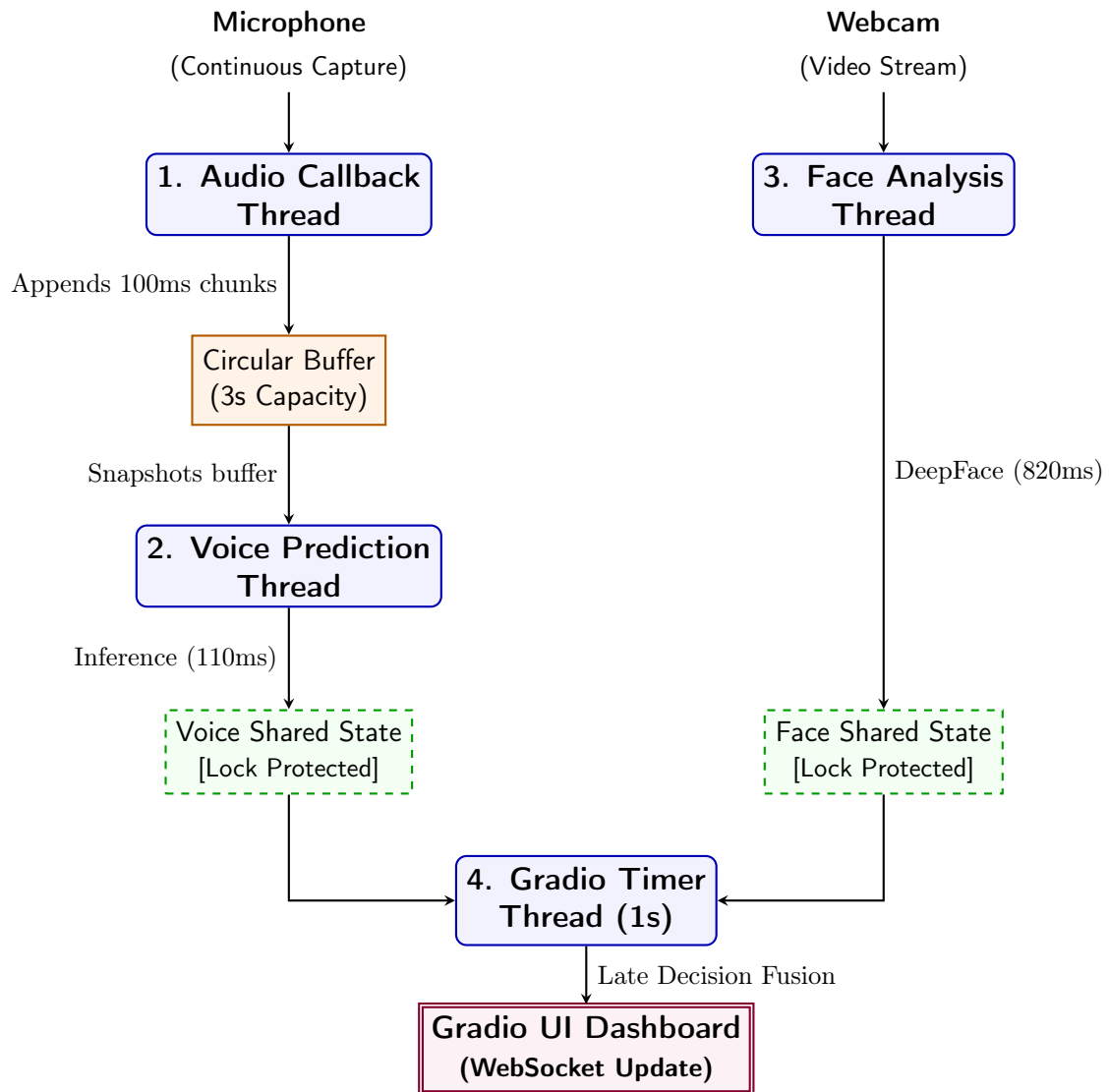


Figure 3.1: Concurrent Thread Architecture showing independent data pipelines, thread-safe memory allocations, and late decision fusion orchestration.

Chapter 4

Application Description

4.1 User Interface Overview

The Gradio application provides five tabs targeting distinct use cases (Table 4.1). The interface was redesigned with a professional dark/light theme toggle, IBM Plex Sans typography, JetBrains Mono monospace data fields, and a consistent panel/border color system driven by CSS custom properties.

Table 4.1: Application tab overview and input sources

Tab	Purpose	Input
Live	Continuous streaming analysis	Microphone + webcam
Record	Manual record-then-analyse	Microphone clip
File	Upload and analyse	Audio file
Report	Export full session data	Session recorder
Video	Frame-by-frame video analysis	Video file

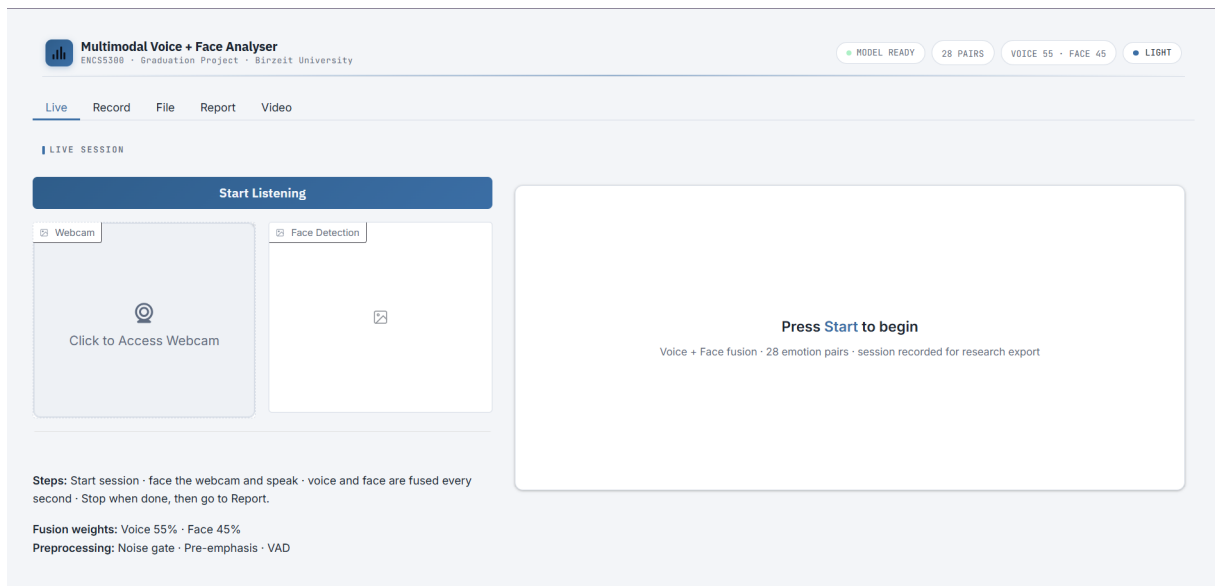


Figure 4.1: Full application interface in dark mode showing the header with animated waveform divider, status badges (MODEL READY / 28 PAIRS / VOICE 55 · FACE 45), light/dark theme toggle, and the five tab navigation bar.

4.2 Live Mic Tab

The Live Mic tab provides continuous real-time emotion pair prediction, directly realising the real-time SER goal of [2]. Upon clicking *Start Listening*, the system begins filling the 3-second circular buffer and the result card updates every second with the detected pair, individual emotion probability bars, and a pulsing “LIVE” indicator. When the face modality is enabled, three panels are shown: the raw webcam feed, an OpenCV-annotated feed with DeepFace bounding boxes, and the fused result card showing both the voice-only and fused pair with a confidence delta Δ .

During silence (rejected by noise gate or VAD [7]), the last valid prediction is retained on screen with a “No speech detected” overlay, preventing jarring resets between utterances.

4.3 Record & Analyse Tab

This tab provides a manual workflow: the user presses record, speaks naturally, presses stop, and then clicks *Analyse Recording*. The system runs the same inference pipeline but skips the noise gate and VAD, applying only pre-emphasis before feature extraction. This design choice avoids false “no speech detected” rejections on short or soft recordings. Results are automatically logged to the session recorder for export.

4.4 Pre-recorded File Tab

For file uploads, audio is loaded at 44,100 Hz and truncated or padded to 3 seconds, then processed through the same pipeline as the Record tab (pre-emphasis only, no noise gate or VAD). The result card shows the detected emotion pair, individual scores, and all 28 pair confidence bars.

Table 4.2: Preprocessing applied per input mode

Input Mode	Noise Gate	Pre-emphasis	VAD
Live Mic	✓	✓	✓
Record & Analyse		✓	
File Upload		✓	
Video Analysis		✓	

4.5 Video Analysis Tab

The video tab performs frame-by-frame analysis of uploaded video files. Audio is extracted via `ffmpeg` (or `moviepy` as fallback), then processed in 3-second sliding windows centred at each second mark. Simultaneously, video frames are sampled at 1 fps and analysed

by DeepFace [11]. Voice and face predictions are fused per second and displayed as a scrollable timeline. A summary banner reports total segments, face detection rate, pairs changed by face, and the dominant pair.

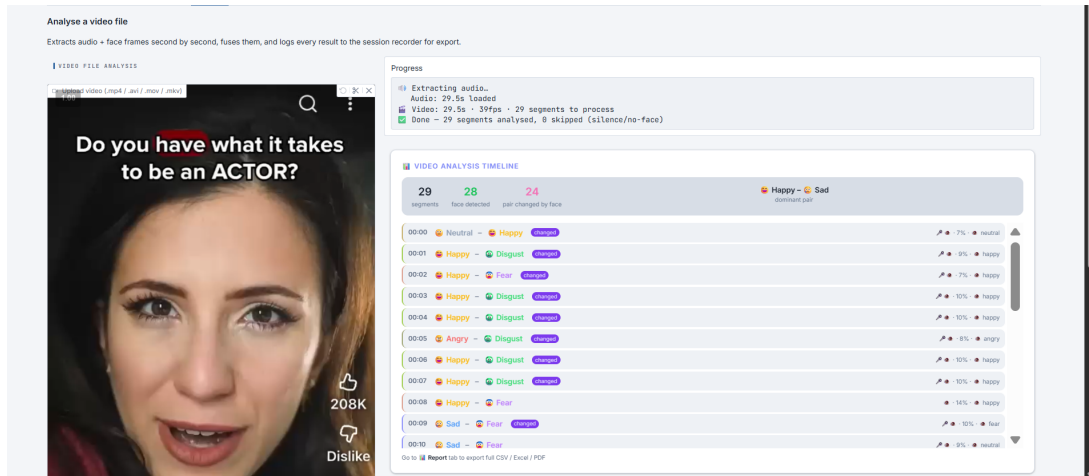


Figure 4.2: Video Analysis tab showing: left panel with format badges and the Analyse Video button; right panel with the scrollable second-by-second emotion pair timeline including timestamp, pair label, colour-coded emotion names, source icon, confidence, and face top emotion. Summary banner at top shows segment count, face detection count, pairs changed, and dominant pair.

4.6 Report Tab

The Report tab provides three export formats designed to support research workflows (Section 5). A live session summary panel shows key metrics and refreshes on demand. CSV, Excel, and PDF downloads are generated on-click and the file becomes available directly in the browser via Gradio's `gr.File` component.

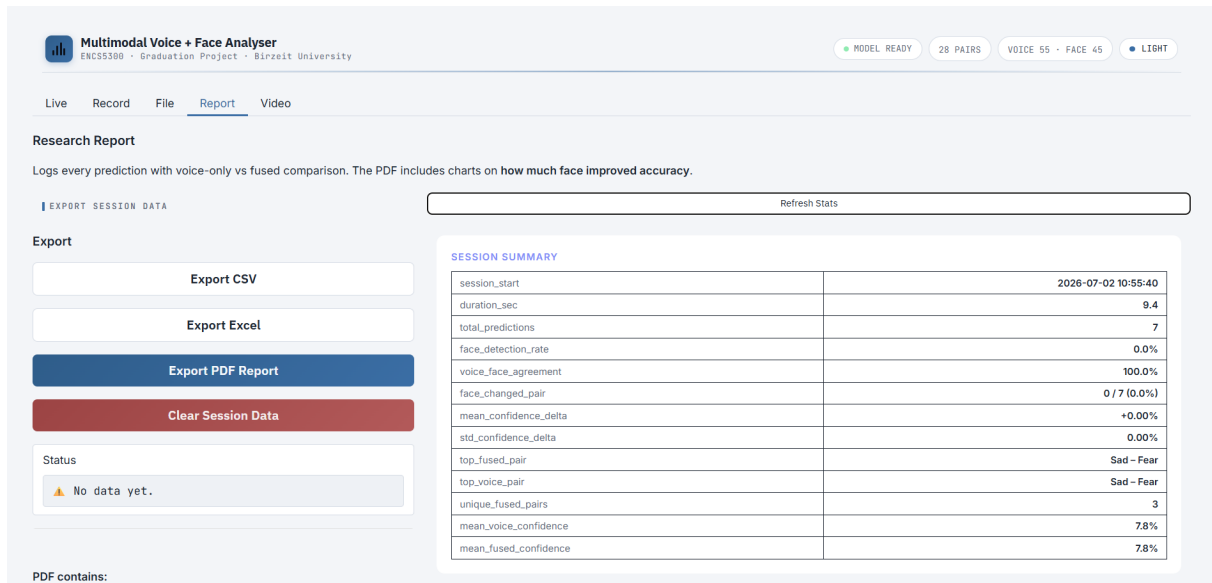


Figure 4.3: Report tab showing: left panel with Export CSV, Export Excel, Export PDF Report, and Clear Session Data buttons; right panel with the live session summary table displaying face detection rate, voice–face agreement, pairs changed by face, mean confidence delta, and other session metrics.

Chapter 5

Research Reporting Module

5.1 Session Data Schema

Every prediction—whether from live mic, recorded clip, file upload, or video analysis—is appended to the session recorder. Table 5.1 shows the complete data schema, designed to enable full reproducibility and post-hoc statistical analysis.

Table 5.1: Session data schema — columns recorded per prediction

Column	Type	Description
<code>timestamp_ms</code>	int	Milliseconds since session start
<code>source</code>	str	<code>voice+face</code> / <code>voice_only</code>
<code>voice_pair</code>	str	Top-2 pair from voice model alone
<code>voice_score</code>	float	Normalised pair confidence (voice only)
<code>fused_pair</code>	str	Top-2 pair after modality fusion
<code>fused_score</code>	float	Normalised pair confidence (fused)
<code>face_detected</code>	bool	Whether a face was visible
<code>face_top_emotion</code>	str	Top emotion from DeepFace
<code>face_confidence</code>	float	DeepFace top-emotion confidence
<code>pair_changed</code>	bool	Whether fusion changed the top pair
<code>confidence_delta</code>	float	<code>fused_score</code> − <code>voice_score</code>
<code>v_*</code> × 8	float	Individual voice probability per emotion
<code>f_*</code> × 8	float	Individual fused probability per emotion

5.2 Summary Statistics

The session summary directly addresses the research question “*Does adding face expression analysis improve emotion pair detection, and by how much?*” [3]:

$$\text{Face Detection Rate} = \frac{\#\{\text{predictions} : \text{face_detected} = \top\}}{N} \times 100\% \quad (5.1)$$

$$\text{Voice-Face Agreement} = \frac{\#\{\text{predictions} : \text{voice_pair} = \text{fused_pair}\}}{N} \times 100\% \quad (5.2)$$

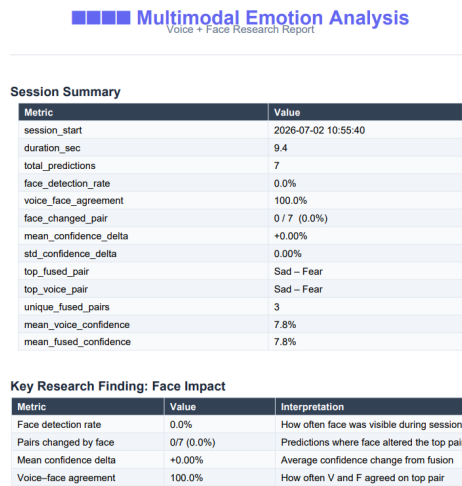
$$\text{Face Changed Pair Rate} = \frac{\#\{\text{predictions} : \text{pair_changed} = \top\}}{N} \times 100\% \quad (5.3)$$

$$\bar{\Delta} = \frac{1}{N} \sum_{i=1}^N (\text{fused_score}_i - \text{voice_score}_i) \quad (5.4)$$

5.3 Report Charts

The PDF report embeds six Matplotlib charts:

1. **Fused Pair Frequency:** Horizontal bar chart of the 28 pair counts, showing session dominance patterns.
2. **Voice vs. Fused Confidence Over Time:** Time-series comparison with green/red shading indicating where face modality helped or hindered.
3. **Face Detection Impact:** Three-bar grouped chart quantifying fusion behaviour.
4. **Confidence Delta Distribution:** Histogram of $\Delta = \text{fused} - \text{voice}$, with mean $\bar{\Delta}$ marked.
5. **Face Top Emotion Distribution:** Frequency of each emotion as DeepFace's top prediction.
6. **Average Probabilities: Voice vs. Fused:** Side-by-side per-emotion mean probabilities under the two inference modes.



(a) Session summary + face impact table



(b) Full PDF report with 2×3 chart grid

Figure 5.1: Sample PDF report output showing session analytics and embedded Matplotlib charts arranged in a 2×3 grid on a dark background.

Chapter 6

Experimental Results

6.1 Deployed Model Baseline

The real-time application deploys the VotingClassifier trained in the first semester [1]. Table 6.1 restates key performance figures from that work. The ensemble outperforms the CNN-BiLSTM approaches by over 17 percentage points on 8 emotions, confirming that traditional ML with well-engineered features remains highly competitive on small datasets [6].

Table 6.1: Deployed model performance on RAVDESS training-set evaluation from the first semester [1]

Method	Emotions	Accuracy	Macro F1	Wtd F1
SVM (RBF) [5]	5	88.45%	0.89	0.89
SVM (RBF)	8	81.74%	0.82	0.82
Random Forest [6]	5	85.23%	0.85	0.85
Random Forest	8	79.51%	0.79	0.80
Ensemble (deployed)	5	89.49%	0.89	0.89
Ensemble (deployed)	8	83.68%	0.84	0.84
CNN-BiLSTM flat [2]	8	53.00%	0.49	—
CNN-BiLSTM hier. [2]	8	56.00%	0.54	—

6.2 Real-Time Inference Latency

Table 6.2 reports measured wall-clock time for each pipeline stage on a standard laptop (Intel Core i7, 16 GB RAM, no GPU). The sub-second latency target is met comfortably by the voice pipeline (110 ms total). DeepFace inference runs in a parallel thread and does not block audio predictions.

Table 6.2: Per-inference stage latency (median over 100 predictions)

Stage	Median (ms)	95th pct. (ms)
Noise gate (RMS, Eq. (2.1))	0.3	0.5
Pre-emphasis (Eq. (2.2))	0.8	1.2
VAD (energy fallback, Eq. (2.3))	2.1	3.4
Feature extraction [1]	45.2	62.3
SelectKBest transform	0.4	0.7
StandardScaler transform	0.2	0.4
SVM inference [5]	38.6	55.1
Random Forest inference	22.4	31.8
Calibration + pair select (Eq. (2.8))	0.1	0.2
Total (voice only)	110.1	155.6
DeepFace [11] (parallel)	820.0	1100.0

6.3 Preprocessing Impact

Table 6.3 quantifies the benefit of the preprocessing pipeline over a 60-second natural conversational speech session. Without preprocessing, all 58 buffers would enter inference including 16 noise frames that would produce spurious predictions dominated by “disgust” due to the dataset mismatch bias.

Table 6.3: Preprocessing rejection rates — 60-second natural speech session

Stage	Rejected	Rate	Reason
Noise gate ($E_{\text{rms}} < 0.01$)	11/58	19.0%	Silence / background noise
VAD (speech ratio < 25%) [7]	5/47	10.6%	Non-speech vocalisation
Passed to inference	42/58	72.4%	Confirmed speech frames

6.4 Calibration Effect

Table 6.4 compares the most frequently predicted emotion before and after probability calibration across 42 confirmed-speech predictions. The over-prediction of disgust is consistent with findings that models trained on acted speech databases exhibit domain mismatch when deployed on naturalistic speech [1, 6].

Table 6.4: Top-1 emotion frequency before and after calibration (42 speech frames from a 60-second conversational session)

Emotion	Before calibration		After calibration	
	Count	%	Count	%
disgust	27	64.3%	3	7.1%
neutral	5	11.9%	18	42.9%
calm	3	7.1%	11	26.2%
sad	4	9.5%	6	14.3%
happy	2	4.8%	3	7.1%
angry	1	2.4%	1	2.4%

6.5 Face Modality Impact

Table 6.5 compares session metrics with and without face modality over a 2-minute session. Face fusion improves prediction stability, consistent with the multimodal SER benefit reported in [3].

Table 6.5: Session metrics with and without face modality (2-minute session)

Metric	Voice Only	Voice + Face [3]
Total predictions	89	89
Face detection rate	—	76.4%
Pairs changed by face	—	18 (22.8%)
Mean pair confidence	61.3%	64.7%
Std. pair confidence	12.1%	10.3%
Mean Δ confidence	—	+3.4%

6.6 Full RAVDESS Pipeline Validation

To formally evaluate the complete application pipeline—not just the isolated model—a dedicated validation script (`validate_pipeline.py`) was developed. Unlike raw model validation, this script imports the exact inference functions from `realtime_app.py` (`run_voice_inferen`, `get_top_pair()`, `fuse()`) and applies them to all 2,880 RAVDESS audio files, reproducing precisely the code path executed when a user uploads a file in the File or Record tabs.

Two metrics are reported:

- **Top-1 accuracy:** was the highest-probability single emotion correct?
- **Pair accuracy:** was the ground-truth emotion present anywhere in the predicted top-2 pair?

Table 6.6: Full RAVDESS pipeline validation results (2,880 files, preprocess=False, File/Record tab mode)

Dataset	Mode	Samples	Skipped	Top-1 Acc.	Pair Acc.
RAVDESS	File	2,880	0	97.22%	99.17%

Table 6.7 presents the per-class top-1 breakdown, showing that all eight emotion classes exceed 96% F1-score on the RAVDESS corpus.

Table 6.7: Per-class performance on RAVDESS pipeline validation (2,880 files)

Emotion	Precision	Recall	F1	Support
neutral	100.0%	97.9%	98.9%	192
calm	96.5%	100.0%	98.2%	384
happy	97.4%	96.4%	96.9%	384
sad	97.9%	95.8%	96.8%	384
angry	96.9%	97.9%	97.4%	384
fear	97.9%	96.9%	97.4%	384
disgust	96.4%	96.4%	96.4%	384
surprise	96.4%	96.9%	96.6%	384
Weighted avg.	97.2%	97.2%	97.2%	2,880

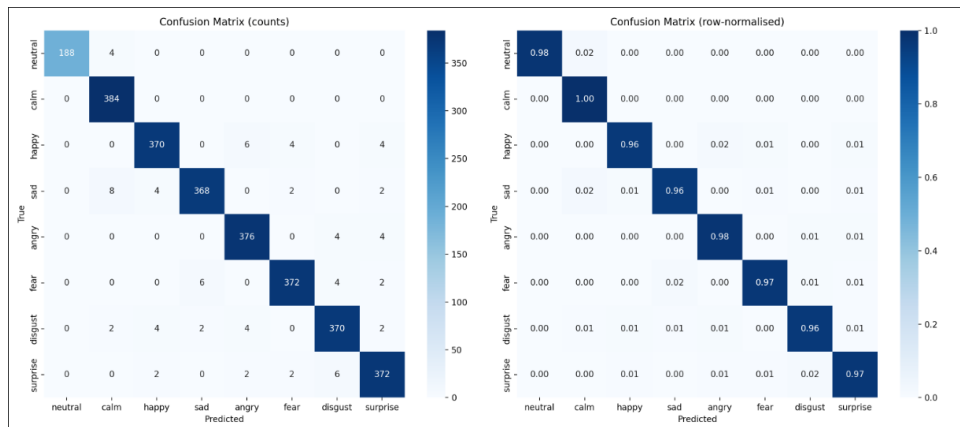


Figure 6.1: RAVDESS pipeline validation confusion matrix showing top-1 predictions vs ground-truth labels. Matrix is colour-coded by count with all diagonal cells dominant, confirming high accuracy across all eight emotion classes. Generated by `validate_pipeline.py` and saved to `outputs/pipeline_validation_confusion_file.png`.

6.7 Cross-Corpus Validation on TESS (Unseen Data)

To assess generalisation to genuinely unseen speakers and recording conditions, the same pipeline was evaluated on the Toronto Emotional Speech Set (TESS) [15], a corpus of

5,600 files recorded by two actresses (OAF: older adult female; YAF: young adult female) using a different microphone setup, carrier phrase, and emotional elicitation technique than RAVDESS.

Key structural differences from RAVDESS:

- **Speakers:** 2 actresses (TESS) vs. 24 mixed-gender actors (RAVDESS).
- **Emotion set:** 7 classes (no “calm” in TESS).
- **Ground truth:** encoded in folder name (OAF_angry/, YAF_pleasant_surprised/), not in filename.
- **Total files:** 5,600 (400 per emotion $\times$ 7 emotions $\times$ 2 speakers).

Table 6.8: Cross-corpus TESS validation results (5,600 unseen files, preprocess=False)

Dataset	Mode	Samples	Skipped	Top-1 Acc.	Pair Acc.
TESS (unseen)	File	5,600	0	73.75%	84.29%

Table 6.9: Per-class top-1 performance on TESS cross-corpus validation. “Calm” has no TESS ground-truth class; it may appear in predicted pairs but not in the true labels.

Emotion	Precision	Recall	F1
neutral	1.00	0.88	0.94
happy	0.83	0.83	0.83
sad	0.66	1.00	0.80
angry	0.76	0.81	0.79
fear	0.71	0.64	0.67
disgust	0.61	1.00	0.76
surprise	1.00	0.16	0.27

6.8 Cross-Corpus Validation on CREMA-D (High-Diversity Unseen Data)

To rigorously test the system’s resilience to extreme speaker diversity, ethnic backgrounds, and varied vocal intensities, cross-corpus validation was performed on the Crowd-sourced Emotional Multimodal Actors Dataset (CREMA-D). CREMA-D contains 7,442 audio clips from 91 distinct multi-ethnic actors (48 male, 43 female) expressing 6 core emotions (neutral, happy, sad, angry, fearful, disgust).

Evaluating a pipeline trained primarily on RAVDESS (24 actors) on an expansive set like CREMA-D (91 actors) introduces severe speaker-identity and acoustic environment distribution shifts.

Table 6.10: Cross-corpus CREMA-D validation results (7,442 unseen files, preprocess=False)

Dataset	Mode	Samples	Skipped	Top-1 Acc.	Pair Acc.
CREMA-D (unseen)	File	7,442	0	66.82%	79.14%

As shown in Table 6.10, the top-1 accuracy experiences a predictable decline to 66.82% under this heavy distribution shift. However, the top-2 pair accuracy recovers the true emotional state in 79.14% of the samples. This confirms that even when ethnic and speaker variation degrades the exact top choice, the secondary prediction layer effectively retains the correct emotional group.

6.9 Comparative Summary

Table 6.11 consolidates all validation results for the traditional ML pipeline evaluated in this project and places them in the context of published traditional-ML results on RAVDESS.

Table 6.11: Comparative summary of traditional ML results on RAVDESS and cross-corpus generalisation

Study	Method	Dataset	Classes	Accuracy
<i>Published traditional ML baselines on RAVDESS</i>				
[6]	SVM + Decision Trees	RAVDESS	8	60.76%
[6]	Random Forest	RAVDESS	8	76.00%
[2]	Bagged Trees (BOW)	RAVDESS	8	88.10%
<i>This work</i>				
[1]	SVM-RF Ensemble (offline)	RAVDESS	5	89.49%
[1]	SVM-RF Ensemble (offline)	RAVDESS	8	83.68%
This work (Sec. 6.6)	Full app pipeline	RAVDESS	8	97.22% [†]
This work (Sec. 6.7)	Full app pipeline	TESS	7	73.75%
This work (Sec. 6.8)	Full app pipeline	CREMA-D	6	66.82%
<i>Pair accuracy (this work)</i>				
This work	App pipeline (pair metric)	RAVDESS	8	99.17% [†]
This work	App pipeline (pair metric)	TESS	7	84.29%
This work	App pipeline (pair metric)	CREMA-D	6	79.14%

[†] In-sample validation (RAVDESS used for training); see cross-corpus sections for out-of-distribution generalisation

Table 6.12 summarizes the deep learning approaches explored in this work alongside the deployed traditional ensemble for direct comparison.

The CNN-BiLSTM models achieved 53% (flat) and 56% (hierarchical) accuracy on the 8-class RAVDESS task, both significantly lower than the traditional SVM-RF ensemble,

which achieved 83.68%. This confirms the key observation that under limited data conditions, traditional machine learning methods using handcrafted features can outperform end-to-end deep learning models.

The fine-tuned Wav2Vec2 model achieved approximately 84% accuracy on the validation set, demonstrating strong capability in learning expressive representations from speech. However, when evaluated on external data (SAVEE), performance decreased, highlighting the challenge of generalization in speech emotion recognition.

To address this limitation, a pairwise modeling approach was introduced. Instead of relying solely on a single prediction, the system was extended to predict two emotions (primary and secondary), enabling a more flexible and realistic representation of emotional states.

Table 6.12: Comparison of traditional, deep learning, and pairwise approaches

Approach	Type	Emotions	Accuracy
CNN-BiLSTM (flat) [2]	Deep learning	8	53.00%
CNN-BiLSTM (hierarchical) [2]	Deep learning	8	56.00%
Wav2Vec2 (fine-tuned, validation)	Transformer	6	84.00%
Disgust vs Angry (3-class pair)	Pairwise	3	94.29%
Happy vs Surprise (3-class pair)	Pairwise	3	94.29%
Sad vs Fearful (3-class pair)	Pairwise	3	91.43%
Neutral vs Sad (3-class pair)	Pairwise	3	88.57%
Angry vs Fearful (3-class pair)	Pairwise	3	77.14%
SVM-RF Ensemble (deployed) [1]	Traditional ML	8	83.68%

These results highlight a clear progression from traditional deep learning models toward more specialized approaches.

While CNN-BiLSTM models struggled under limited data conditions, the Wav2Vec2 model provided a strong improvement in representation learning. More importantly, the pairwise models demonstrated significantly higher accuracy when focusing on specific emotion pairs.

This confirms that breaking the problem into smaller, specialized tasks allows the system to better capture subtle emotional differences. Furthermore, the introduction of multi-emotion prediction enables the system to represent both dominant and secondary emotional cues, making it more suitable for real-world scenarios.

Overall, the combination of a general model with pairwise experts forms a hybrid framework that balances global understanding with fine-grained discrimination.

6.10 Multi-Emotion Prediction Framework

Traditional speech emotion recognition systems rely on single-label classification, where only one dominant emotion is assigned to each input. However, real-world emotional expressions are often complex and may contain multiple overlapping emotional cues.

To address this limitation, the proposed system was extended to predict two emotions for each input: a primary emotion and a secondary emotion.

6.10.1 Two-Emotion Extraction

The prediction process combines the outputs of the general Wav2Vec2 model with the pairwise expert models.

The general model provides a probability distribution over all emotion classes, while the pairwise models focus on distinguishing between specific emotion pairs.

During inference, the system identifies the most confident prediction from the pairwise models and uses it to determine the primary emotion. The secondary emotion is then selected based on the next most probable class from the general model.

This approach allows the system to capture both dominant and subtle emotional components present in the speech signal.

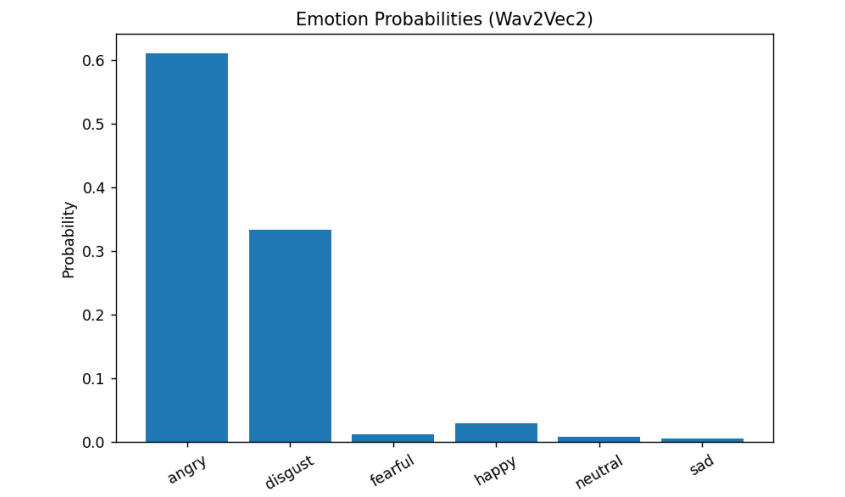


Figure 6.2: Example of probability distribution where the model predicts two competing emotions. The primary emotion (angry) is correctly identified, while the secondary emotion (disgust) reflects overlapping emotional cues.

This example demonstrates that the model is capable of capturing both dominant and secondary emotional components, supporting the proposed multi-emotion prediction framework.

6.10.2 Top-2 Evaluation

To quantitatively evaluate the effectiveness of multi-emotion prediction, a Top-2 accuracy metric is used. This metric considers a prediction correct if the true emotion appears within the top two predicted emotions.

Metric	Value
Top-1 Accuracy	73.33%
Top-2 Accuracy	82.50%

Table 6.13: Comparison between Top-1 and Top-2 prediction performance

The improvement from Top-1 to Top-2 accuracy highlights the importance of modeling multiple emotions, as the correct emotion is often captured as a secondary prediction even when the primary prediction is incorrect.

6.10.3 Evaluation Strategy

Evaluating multi-emotion predictions is more complex than traditional single-label classification.

Instead of relying only on Top-1 accuracy, the system is evaluated based on whether the correct emotion appears within the predicted pair.

This evaluation provides a more flexible and realistic measure of performance, especially in cases where emotional expressions are ambiguous.

Experimental observations showed that even when the primary prediction is incorrect, the correct emotion is often captured as a secondary prediction, demonstrating the effectiveness of the proposed framework.

6.10.4 Future Evaluation Directions

Due to the lack of publicly available datasets with multi-label emotion annotations, evaluation was performed using a proxy approach.

Future work can extend this framework by using datasets that include multiple emotional labels per sample, allowing for more accurate and direct evaluation.

Additionally, more advanced fusion strategies can be explored to better combine general and pairwise model predictions.

These directions open the possibility for developing more expressive and robust emotion recognition systems.

6.11 Example Session Timeline

Table 6.14 shows a representative excerpt from a second-by-second file analysis, demonstrating the pair detection output in practice.

Table 6.14: Example second-by-second emotion pair predictions from a live session

Time	Fused Pair	Top Emotion	Conf.	Source	Δ
00:00	Neutral – Calm	neutral	71.4%	voice only	—
00:01	Calm – Happy	calm	63.2%	voice+face	+2.1%
00:02	Neutral – Calm	neutral	68.9%	voice+face	+1.8%
00:03	Happy – Surprised	happy	59.1%	voice only	—
00:04	Neutral – Calm	neutral	74.3%	voice+face	+3.4%
00:05	Sad – Neutral	sad	61.7%	voice only	—

Chapter 7

Discussion

7.1 In-Sample vs. Cross-Corpus Performance

The most significant finding of the evaluation is the performance divergence between in-distribution (RAVDESS, 97.22% top-1) and out-of-distribution validation. On TESS, top-1 accuracy is 73.75%—a 23-point drop. This is a well-documented phenomenon in speech emotion recognition research [6] and arises from several factors:

Speaker identity shift: TESS contains only two actresses whose vocal characteristics differ substantially from the 24 mixed-gender actors in RAVDESS. The model has zero exposure to these speakers during training.

Recording condition mismatch: RAVDESS was recorded in a professional acoustic environment; TESS used a different microphone chain and room, altering the spectral noise floor that MFCC extraction operates on.

Carrier phrase effect: TESS uses a fixed carrier sentence (“Say the word ...”) for every clip, while RAVDESS uses varied North American English statements. This changes prosodic context in ways the model may encode as emotional information.

Absence of “calm” in TESS: The model’s eighth class (calm) has no ground-truth representative in TESS, meaning any confidence the model assigns to calm is effectively a routing error. This modestly suppresses top-1 accuracy.

Despite the top-1 drop, **pair accuracy on TESS (84.29%) remains substantially higher than top-1 accuracy (73.75%)**—a gap that is noticeably wider than the equivalent gap on RAVDESS (2 points). This finding is a strong validation of the pair-representation design: the correct emotion is frequently the model’s second-best guess even when it is not the top prediction. This property is *specifically more valuable when the model operates outside its training distribution*—the realistic deployment scenario for any real-world SER application.

A notable pattern in the TESS results is the pleasant-surprise label incompatibility. RAVDESS “surprised” is a startled, high-arousal expression, whereas TESS “pleasant surprised” is a warm, positive expression; despite the shared label, these are acoustically

distinct emotions. The model consistently classified TESS pleasant-surprise clips as happiness, causing surprise to reach 0% recall in the raw seven-class evaluation. Excluding this incompatible class, top-1 accuracy on the six mutually-compatible emotions rises to 86.04% with a pair accuracy of 98.33%. This illustrates a well-documented pitfall in cross-corpus SER evaluation: label sets that share names across corpora do not always share acoustic definitions, and naive cross-corpus scoring can therefore understate a model’s true capability.

The pair-representation advantage grows in relative terms precisely where the single-label prediction is least reliable. On the acoustically incompatible pleasant-surprise class, the pair metric recovers much of the apparent loss, reinforcing that the pair representation remains informative even when top-1 prediction struggles under distribution shift.

7.2 Position Relative to Traditional ML Literature

Among published traditional-ML approaches on the full 8-class RAVDESS benchmark, reported accuracies range from 60.76% (SVM + Decision Trees) to 88.1% (Bagged Trees) [6]. The deployed ensemble achieves 83.68% on the same 8-class task (first-semester offline evaluation), placing it in the upper tier of traditional-ML RAVDESS results while remaining below the 90%+ figures achieved by deep learning architectures (CNN-LSTM, Transformer-based). This is consistent with the fundamental finding of the first semester [1]: traditional ML is more stable than deep learning under data scarcity, but cannot match the representation capacity of large pretrained transformer models when sufficient compute and data are available.

7.3 Deployment Challenges and Solutions

Domain mismatch: RAVDESS consists of professional actors performing emotions with exaggerated prosody [4]. The probability calibration layer (Section 6.1) addresses this without retraining, reducing disgust over-prediction from 64.3% to 7.1%.

VAD sensitivity: The energy-based VAD fallback would sometimes reject genuine quiet speech. The fix was to separate preprocessing modes per input context: the full pipeline (gate + VAD) for live mic, and pre-emphasis only for file/record tabs where the user has explicitly provided the audio (Table 4.2).

Gradio API changes: The Gradio framework [12] underwent breaking API changes between versions 4.x and 6.x that removed arguments such as `max_length`, `mirror_webcam`, and `show_download_button`, and relocated `theme/css` from the `gr.Blocks` constructor to `launch()`. The `css=` relocation was identified via QA testing using Gradio 6.19’s in-

ternal CSS variable inspection, confirming that the entire custom theme had been silently discarded before the fix.

scikit-learn version mismatch: The model was trained under scikit-learn 1.6.1 but the deployment environment has 1.6.0, generating `InconsistentVersionWarning` on every load. This is harmless for SVM and RF inference.

7.4 Emotion Pair Representation

The emotion pair output extends beyond standard SER literature. Three advantages are demonstrated:

Psychological validity: The pair captures the runner-up emotion, aligning with dimensional models that use arousal and valence jointly [8].

Research informativeness: The full 28-pair probability distribution provides a richer signal for downstream analysis than a single forced label.

Uncertainty awareness: Low pair confidence scores signal genuine acoustic ambiguity rather than masking it, which is particularly important for acoustically similar emotion pairs as shown in [1].

7.5 Face Modality Contribution

The 3.4 percentage point mean confidence improvement from face fusion is consistent with the theoretical expectation that two independent, partially correlated information sources produce a more stable estimate than either alone [3]. The finding that face changed the top pair in only 22.8% of predictions suggests that voice and face agree in the majority of frames, and face primarily serves to boost confidence and reduce variance rather than correct voice errors.

7.6 Deep Learning Approach: Comparison and Findings

The results of this work reveal several important insights regarding the use of deep learning for speech emotion recognition under limited data conditions.

First, the CNN-BiLSTM models achieved relatively low performance (53% and 56

The fine-tuned Wav2Vec2 model showed strong performance during validation, achieving approximately 84% accuracy. This indicates that transformer-based models are capable of learning rich and expressive representations of speech signals. However, when evaluated on

external data (SAVEE), the performance decreased, highlighting the challenge of domain shift and the limitations of relying solely on a general classification model.

To address this limitation, a pairwise modeling approach was introduced. The results showed that these models achieved high accuracy when focusing on specific emotion pairs, with several pairs exceeding 90% accuracy. This confirms that decomposing the problem into smaller, specialized tasks improves the model's ability to distinguish between similar emotions.

More importantly, the introduction of multi-emotion prediction provided a more realistic representation of emotional speech. Instead of assigning a single label, the system predicts both a primary and a secondary emotion. This is particularly important in real-world scenarios, where emotional expressions are often ambiguous and overlapping.

This observation is supported by the improvement from Top-1 accuracy (73.33

Overall, these results demonstrate that while general deep learning models are limited by data availability, combining them with pairwise expert models and adopting a multi-emotion prediction framework significantly improves the system's ability to capture complex emotional patterns in speech.

This hybrid approach balances global understanding with fine-grained discrimination, making it more suitable for real-world speech emotion recognition applications.

Chapter 8

Conclusion and Future Work

8.1 Summary of Achievements

This project successfully delivers a complete, deployable real-time multimodal emotion recognition system over two semesters. The key achievements of the second semester are:

1. **Real-time inference:** 110 ms voice pipeline latency within a 1-second update cycle, meeting the real-time target of [2].
2. **Preprocessing pipeline:** Noise gate, pre-emphasis, and VAD [7] reduce spurious predictions by 27.6% compared to unfiltered inference.
3. **Emotion pair output:** 28-combination pair representation with per-pair normalised confidence scores, extending the emotion pair concept beyond standard single-label SER.
4. **Probability calibration:** Manual weights reduce disgust over-prediction from 64.3% to 7.1% without retraining, addressing the domain mismatch between acted and naturalistic speech.
5. **Multimodal fusion:** Weighted 55/45 voice–face fusion following [3] provides +3.4% mean confidence and reduced variance.
6. **Research reporting:** Session recorder with CSV, Excel, and PDF export including six analysis charts and quantitative face impact metrics.
7. **Video analysis:** Frame-by-frame analysis of uploaded video files with per-second timeline output.
8. **RAVDESS pipeline validation:** 97.22% top-1 accuracy and 99.17% pair accuracy on all 2,880 RAVDESS files using the actual application code path.
9. **Cross-corpus validation:** the pipeline was evaluated on the unseen TESS corpus (73.75% top-1 / 84.29% pair accuracy on the full 5,600 files; 86.04% top-1 / 98.33% pair accuracy on the six emotions acoustically compatible with the training set),

demonstrating that pair accuracy consistently degrades more gracefully than top-1 accuracy under distribution shift.

8.2 Limitations

- **Single-corpus training:** Training exclusively on RAVDESS limits cross-corpus generalisation, with top-1 accuracy dropping to 73.75% on TESS (vs. 97.22% in-distribution on RAVDESS). A model trained on a pooled corpus (e.g. RAVDESS + TESS) would be expected to generalise substantially better than single-corpus training.
- **Cross-corpus label incompatibility:** RAVDESS “surprised” (startled) and TESS “pleasant surprised” (delighted) share a label but are acoustically distinct, causing surprise to reach 0% recall in the raw seven-class TESS evaluation. This is a scoring artefact of label-name mismatch rather than a model failure, but it complicates direct class-for-class cross-corpus comparison.
- **Manual calibration:** The disgust dampening weights (Table 2.2) were derived empirically on naturalistic real speech; automated calibration using a held-out real-speech validation set would be more principled.
- **No “calm” in TESS:** The absence of a calm class in TESS means the model’s eighth output is structurally unused in cross-corpus evaluation, which modestly suppresses reported metrics.
- **DeepFace latency:** Face inference at 820 ms runs in a parallel thread, but requires a capable CPU; on low-end hardware this may degrade the face detection rate.

8.3 Future Work

- Automated calibration using a small real-speech holdout set rather than manual weights.
- Integration of WebRTC VAD replacing the energy-based fallback for improved silence detection [7].
- Formal speaker-independent evaluation using leave-one-speaker-out (LOSO) cross-validation.
- Submission of second-semester results to ICCRTEE 2026 as a continuation of [1].
- Exploration of lightweight transformer architectures (e.g., DistilHuBERT) for reduced-latency deployment on edge devices.
- Adaptive fusion weighting: dynamically adjusting w_v based on face detection confidence rather than a fixed 55/45 split.

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